Particles from Native Overlap Transport: Common Currents, Defect Modes and a Pixel-Dependent Mass Operator

B. Müller Mario Poneder Alexander Osika Kai Xue Daniel Ari Friedman

November 7, 2026

Paper release: r2062 **Released:** November 7, 2026

Author affiliations: Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks; Daniel Ari Friedman, Active Inference Institute and FractiAI.

Abstract

Observer Patch Holography proposes that particles are reproducible features of the records of one specified observer, repair and memory process, and that stable massive matter consists of localized defects that repair cannot erase. We give the particle content of that process and compare it with measurement. Complete-response premises fix the Standard Model gauge type, and a declared lift gives one chiral generation. On a specified interacting cube, exact enclosures give 228 charged lines visible to single-fermion probes and 138 composite lines invisible to them, and a compiled instrument detects the composite response. The same localized modes carry all twelve native currents and a pixel-dependent charged-lepton matrix realized by complete memory dynamics. A compact opposite-winding texture has finite energy and four isolated modes, and its quartic extension has exact interacting poles. The specified mass branch gives twelve comparison values. Charged leptons lie within 71 parts per million under a pole premise and a target-informed law; the W, Z, Higgs and top lie within one percent; five lighter quarks miss by 23 to 209 percent. All twelve lie outside their printed intervals. We prove the exclusions this comparison supports and list what a source-selected spectrum must supply.

1 The particle question

Observer Patch Holography (OPH) proposes a self-referential complete history: observers infer the specification of their world from shared records and realize it through operations within that same history. The executable version specifies a carrier Hilbert space, accessible algebras, repair instruments, a scheduler, memory, feedback and preparation services [2, 6]. These are permitted design choices. Operational agreement concerns accepted shared data; state selection and endogenous implementation of response transformations are additional stated principles, recorded with the exact theorem premises in the common registry [25].

The source specification is a tuple

$$\mathcal{S} = (\mathcal{H}, \mathcal{A}, \{\Phi_{e,r}\}, \sigma, \mathcal{M}, \mathcal{F}, \mathcal{P}),$$

where $\Phi_{e,r}$ is the operation on seam e with retained outcome r, σ selects operations, $\mathcal{M}$ is memory, $\mathcal{F}$ is feedback, and $\mathcal{P}$ is the preparation domain. A particle assertion concerns the joint and

conditional laws of actual preparations, interventions and detector outputs of $\mathcal{S}$. Two equal spectra do not identify the states, currents, clocks or instruments that produced them, and Section 7 proves this in several exact forms.

Hypothesis 1.1 (Matter as localized defects)**.** *Stable massive matter admits a localized excitation of a continuing source reference, with an invariant preserved by the admitted dynamics. Its creation, transport, interactions and detection are operations of the same source. Its total excess energy and propagation use one energy and clock dictionary. Unstable species may be metastable or internal excitations; massless radiation need not carry a defect invariant.*

The hypothesis does not ask for a distinct conserved topological charge on every mass eigenstate. Flavor mixing and decay can connect internal states of a defect family, and a protected label cannot change at an admitted vertex without another field or record carrying the compensating charge. Section 2.4 gives the exact constraint.

1.1 Main results

Table 1 states what this paper establishes. Three kinds of statement occur and we keep them apart. Some results follow from the unchanged source with every outcome retained. Some are exact on a chosen design, whose inputs are stated. Some compare a specified branch with measurement, and these are conditional postdictions.

Table 1: Main results and the inputs each one consumes.

Result	**Inputs**
Internal charges of one chiral generation, three Higgs lines and the conditional quadratic mass table (Section 2)	Complete response, inner rotations, spin lift, three family copies, canonical kinetic forms
Complete transfer spectrum of the native carrier, every compact core among 4096 port sets, a positive-excess phase excitation and a positive distance from every quadratic (Section 3.1)	Uniform seam selection and the fair coin; no fitted coefficient
Certified charged spectra of the interacting cube, 228 linear lines and 138 composite lines, and a compiled composite detector (Section 3.2)	Couplings $c = 1/8$, $U = 1$; imported Lieb uniqueness; pulse services
Complete classification of conserved quadratics (Theorem 3.1)	Open connected box, positive hopping and repulsion
A complete memory realization of the charged coefficients and certified pixel feedback roots (Section 4)	Target-informed mass law; declared probes, gain and scale design
One localized algebra for currents and masses, a compact finite-energy pair, a gauge-covariant paired band, exact interacting poles and a controlled work operator (Section 5)	Chosen textures, coherent pairing service and quartic coupling
Twelve comparison values against the 2026 Particle Data Group tables (Section 6)	Target-informed branch, external unit, supersymmetric running proxy, imposed criticality
Exact exclusions of mass attachments and record-based selection rules (Section 7)	Each states its own domain

No result in this paper derives a measured particle mass, and the construction is a precise conditional particle model. Section 8 lists what a source-selected spectrum must supply.

Contents

1 The particle question **1**
1.1 Main results . . . 2

2 Internal quantum numbers **4**
2.1 Complete response and the compact Lie algebra . . . 4
2.2 The matter module and its family factor . . . 4
2.3 The conditional field action . . . 5
2.4 Flavor, neutrinos and conserved defect labels . . . 6

3 The spectrum of the native source **6**
3.1 Transfer modes, compact cores and a phase excitation . . . 6
3.2 The charged spectrum of the interacting cube . . . 7
3.3 Complete conserved quadratics . . . 8
3.4 One-body massless and massive sectors . . . 9

4 The charged mass operator **9**
4.1 Pixel capacity and the mass law . . . 9
4.2 A pixel mass law in native memory dynamics . . . 10
4.3 Memory correlations beyond the mean unitary . . . 12
4.4 Complete records, capacity and recovery . . . 12
4.5 Flavor from current-gated memory . . . 13
4.6 Pixel feedback with one charged-fermion mass prescription . . . 13

5 Localizing the same current and mass operators **14**
5.1 Topology, total energy and the radial source . . . 14
5.2 One local carrier for defect states, currents and mass coefficients . . . 15
5.3 Compact opposite-degree textures and their isolated modes . . . 17
5.4 Protected texture records and controlled band transport . . . 18
5.5 Gauge-covariant paired compact bands . . . 19
5.6 A quartic interaction on the complete compact band . . . 19
5.7 Work on the moving compact interacting band . . . 22

6 Comparison with the measured spectrum **23**
6.1 Observables and conventions . . . 23
6.2 The twelve comparison values . . . 24
6.3 Charged leptons . . . 25
6.4 Quarks . . . 25
6.5 Weak bosons, Higgs and top . . . 25
6.6 Massless and neutral sectors . . . 26

7 What records and symmetries exclude **26**
7.1 Coherent phases and family averaging . . . 26
7.2 The passive family metric . . . 27
7.3 Neutral hierarchy . . . 27
7.4 Residues, references and mass readouts . . . 27
7.5 Comparing capacities . . . 28

8 The catalogue and what a source-selected spectrum must supply **28**
8.1 What the displayed source assigns 28
8.2 The required attachments 29
8.3 Interaction, chiral continuum and responsive geometry 30

2 Internal quantum numbers

2.1 Complete response and the compact Lie algebra

The carrier is the oriented icosahedral boundary, with twelve ports, thirty seams and twenty faces. Its real port module under the proper rotation group A_5 decomposes as

$$V_{12} = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}, \qquad \dim V_{12}^{A_5} = 1.$$

Assume a faithful real-linear response map $D : V_{12} \to \mathfrak{u}(\mathcal{K})$ whose image is the complete connected response algebra and is closed under commutators, with proper carrier rotations acting through the same connected inner response group up to operationally silent centralizer factors. The gauge paper [5] proves that the compact classification then gives

$$\mathfrak{g} \simeq \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3), \tag{1}$$

because inner rotations fix a centre of dimension at most one and a centre-free $\mathfrak{su}(2)^4$ cannot have a one-dimensional fixed space, which leaves the semisimple split $11 = 3 + 8$. It also constructs an explicit current law on the protected native carrier realizing $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Twelve ports alone do not imply (1), and carrier rechartings are distinct from physical spatial rotations of a relativistic scattering theory. This paper consumes the result and its twelve current matrices.

2.2 The matter module and its family factor

A declared spin lift to five internal coordinates and the nonvacuum even exterior module give

$$M_{15} = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5, \qquad \dim M_{15} = 10 + 5 = 15,$$

the familiar $\mathbf{10} \oplus \overline{\mathbf{5}}$ branching [9]. The source-current lift preserves all brackets and has an exact intertwiner with this module. Its simultaneous primitive-charge and color/weak Casimir readouts give the following all-left convention.

Label	Color	Weak	$6Y$	Components
Q	$\mathbf{3}$	$\mathbf{2}$	1	6
u^c	$\overline{\mathbf{3}}$	$\mathbf{1}$	-4	3
d^c	$\overline{\mathbf{3}}$	$\mathbf{1}$	2	3
L	$\mathbf{1}$	$\mathbf{2}$	-3	2
e^c	$\mathbf{1}$	$\mathbf{1}$	6	1

The labels abbreviate computed finite quantum numbers; their identification with laboratory Weyl fields requires the physical current, spin and kinetic maps. The gauge paper [5] gives the anomaly scan, which selects this module among exterior masks up to conjugation, the $\mathbb{Z}_6$ global form and the conditional family theorem. For the particle construction we explicitly choose $M_{15} \otimes \mathbb{C}^3$, with gauge currents acting trivially on $\mathbb{C}^3$. A two-dimensional radial kernel or a winding number

three is no proof of three generations, and equivalent triplet representations on spatial, color and family factors do not identify those factors.

Exchange records have a separate finite route. On the native ordered-cover transport, let $p_h(x)$ be the permutation parity accumulated by branch h. A coherent ordering bit gives

$$K_h = \frac{1}{\sqrt{2}} \sum_{x \in D_h} |f_h(x)\rangle\langle x| \otimes X^{p_h(x)}, \qquad Q = I \otimes X, \qquad [K_h, Q] = 0.$$

An odd exchange loop acts as X on the bit, and the corresponding colored fermionic lift acts as $-$SWAP. The sectors survive complete record-adaptive native schedules in this specified memory design; dephasing or resetting the bit changes that conclusion. These are exchange-history sectors, distinct from the spatially localized texture charge of Section 5.

2.3 The conditional field action

Choose one complex weak doublet H of hypercharge $1/2$, canonical positive kinetic forms, physical Weyl statistics and a Lorentz-covariant local field interpretation in $3+1$ dimensions. Restrict the polynomial action to operator dimension at most four, modulo integration by parts. The invariant grammar then gives the gauge kinetic terms, scalar kinetic term and potential, fermion kinetic terms and exactly three charged Yukawa lines. Independent gauge couplings, topological theta terms and an additive vacuum-energy constant are also free. In the four-component convention the charged Yukawa part is

$$\mathcal{L}_Y = -\overline{Q}_L \widetilde{H} Y_u u_R - \overline{Q}_L H Y_d d_R - \overline{L}_L H Y_e e_R + \text{h.c.}$$

Each Y_a is a complex 3×3 matrix. The native invariant-line computation solves all twelve current equations for both Higgs components and gives one line for each allowed origin $(6Y_1, 6Y_2) = (-4, 1), (1, 2), (-3, 6)$. Tensoring these lines with the gauge-trivial family factor leaves exactly the three matrix coefficients. Canonical kinetic normalization must precede any reading of their singular values as action masses [5, 24, 23].

For the explicitly chosen broken potential

$$V(H) = \lambda_H (H^\dagger H - v^2/2)^2, \qquad \lambda_H > 0,$$

the vacuum shell has $|H| = v/\sqrt{2}$. Taking $H = (0, v/\sqrt{2})$ gives the following conditional quadratic table.

Field directions	Action mass or coefficient
Three charged leptons	$m_{e,i} = v\, s_i(Y_e)/\sqrt{2}$
Three up and three down flavors	$m_{a,i} = v\, s_i(Y_a)/\sqrt{2}$, each in three colors
Three minimal left neutrino fields	Zero in the dimension-four charged grammar
Charged weak vectors	$m_W = g_2 v/2$
Neutral massive weak vector	$m_Z = v\sqrt{g_2^2 + g_Y^2}/2$
Electromagnetic direction	Zero quadratic mass
Eight color directions	Zero quadratic mass; no free physical gluon claim
One radial scalar	$m_h = \sqrt{2\lambda_H}\, v$

The neutral vector matrix follows from the coefficient of $\frac{1}{2}A^T M^2 A$ in $|DH|^2$; this factor of two matters. The three zero scalar Hessian directions are gauge-orbit directions. Neither the sign of the broken potential nor its quantum vacuum is selected by the finite representation.

For a supplied Weyl principal symbol and the antisymmetric pairing $\epsilon_{\mathrm{spin}} \otimes \overline{K}$, squaring the complete Nambu symbol cancels its momentum–mass cross terms, so

$$E_j(\mathbf{p})^2 = c^2|\mathbf{p}|^2 + s_j(K)^2.$$

This is a theorem of that supplied relativistic action. The spatial source of Section 5 has additive bands instead, and the formula does not transfer to it by matching K alone.

2.4 Flavor, neutrinos and conserved defect labels

The canonical family-frame freedom is $\mathrm{U}(3)^5$, and it classifies every same-current, canonical-kinetic family embedding. It preserves the charged singular spectra and basis-invariant mixing data. In particular $V_{\mathrm{CKM}} = U_{uL}^\dagger U_{dL}$ is determined by the same left-handed weak current; relabeling families cannot repair a mass hierarchy or generate physical CP violation. A real symmetric coefficient class has $\mathrm{tr}[Y_u Y_u^\dagger, Y_d Y_d^\dagger]^3 = 0$ [22].

The native current also supplies one LH invariant line, and its symmetric family lift enters the complete action with six free complex Weinberg coefficients. A supplied dimension-five operator $(L \cdot H)^T C_5 (L \cdot H)/(2\Lambda) + \mathrm{h.c.}$ gives $M_\nu = v^2 C_5/(2\Lambda)$ and Takagi singular masses. A calibrated instance has one massless neutrino and two nonzero coefficients. The charged-lepton construction supplies neither C_5 nor its scale.

Conserved additive defect labels must respect the same vertices. If the charged-current mixing graph is connected, the relation

$$q(W^-) - q(e_\alpha) + q(\nu_i) = 0$$

for every nonzero entry forces one common neutrino label N and one common charged-lepton label. A massive Majorana field adds $2N = 0$, which in an integer winding group implies $N = 0$. If a massless neutrino is excluded from defect assignments and has label zero, connected mixing forces the other neutrino labels to zero as well. Distinct protected winding numbers therefore cannot simply be assigned to the three mixed mass eigenstates. Internal excitations, nonadditive invariants and charged compensating vertices require their own source construction.

3 The spectrum of the native source

Before any mass law is chosen, the source has spectra of its own. This section collects them. No coefficient here is fitted to a particle, every energy is in the source's own dimensionless unit, and each result retains the complete state space on which it is stated.

The native source acts on twelve nonnegative integer port loads of total two, which form 78 configurations with an invariant shell of 66 binary ones. For an oriented seam (a, b) with contrast $d = x_a - x_b$ and a fair placement coin c, repair moves $j = \frac{1}{2}[d - (2c-1)(|d| \bmod 2)]$ units from a to b. Instrument C records (c, j) and instrument E also records the seam-total parity; they share one classical update and differ in the coherence they retain [6].

3.1 Transfer modes, compact cores and a phase excitation

Keep all sixty-six binary total-two configurations. Uniform independent seam selection and fair placement give the complete attempt transfer $P = C/60$, with $2I/3 \le P \le I$, from all 3960

endpoint branches. Exact characteristic polynomial factorization, independently checked by sixty-six Newton traces, finds sixteen distinct eigenvalues and a unique constant ground vector. The largest nontrivial eigenvalue is $(55+\sqrt{5})/60$.

A compact amplitude core means that both loads lie in a selected port set. The largest confined invariant subspace is the kernel of the outgoing block and all its products with powers of the internal block. Exhausting all 4096 subsets, with exact rank certificates, gives no nonzero such subspace on nine or fewer ports. Six ten-port cores support three modes, with transfer eigenvalues 4/5, 5/6, 13/15 and integrated normalized native autocorrelations 5, 6, 15/2 attempts. Admitted changes of seam weights destroy their confinement, and the full recorded instruments do not conserve their transfer sector. A common invariant amplitude subspace under every native seam uses all twelve ports.

The faithful reconstructed reference $\rho = P/\operatorname{Tr} P$, with $\operatorname{Tr} P = 56$, has a different localized excitation. Let $D = (-1)^{n_0}$ and $\rho' = D\rho D$. It has the same entropy, configuration probabilities and full eleven-port exterior as ρ. For the source-derived $H = -\log P$,

$$\frac{1}{1008} \le \operatorname{Tr}(H(\rho' - \rho)) \le \frac{1}{672}.$$

The bounds follow from $\|[P, D]\|_{\mathrm{HS}}^2 = 1/9$ and the positive spectral bounds on P. The native C and E unread channels contract the state contrast by 5/6 and 7/10 per uniform attempt, and complete tape-plus-exterior laws agree when compared with the equally evolved reference. The excitation is the reconstructed functional $A \mapsto \ell(DAD)$ and a decaying finite excitation; no native phase gate or coherent preparation is inferred from it.

Interaction is detectable without a fitted spectrum. Every quadratic operator in this fixed occupation dictionary has diagonal $K_{\{i,j\},\{i,j\}} = c + \epsilon_i + \epsilon_j$, regardless of its hopping terms, so its four-pair contrast on $01, 23, 02, 13$ vanishes. For $I - P$ the same contrast is $-1/15$, which puts it at norm distance at least $1/60$ from every such quadratic. A positive logarithm expansion with an explicit tail bounds the distance for $-\log P$ below by 0.01984562. The test retains every native state and permits arbitrary quadratic hoppings, on-site potentials and an energy offset. Arbitrary re-encodings lie outside it.

3.2 The charged spectrum of the interacting cube

The open eight-site Weyl/contact cube is the smallest interacting source with a complete charged spectrum. In its exact hidden-spin frame the source is

$$K = \mathrm{d}\Gamma(A \otimes I_2)/16 + \sum_x n_{x\uparrow} n_{x\downarrow} - N/2, \qquad A = A^T, \quad A^2 = 3I,$$

at $c = 1/8$, $U = 1$. All source matrices come from the full 65,536-state integer Hamiltonian. The reference is the unique half-filled ground, with hidden spin zero, eta zero and energy in $[-4.12405873337830, -4.12405873337828]$. Its uniqueness uses the theorem of Lieb on the half-filled Hubbard model [10] together with a full-number-sector bound. Hidden spin and eta are internal algebras, and neither carries a Lorentz-spin assignment.

Lines visible to single-fermion probes. Every local one-spin addition or removal measure has exactly 228 distinct positive energies, each with rigorously positive weight. The signed retarded measure has 456 atoms, each charge branch has total weight 1/2, and every eight-site residue matrix has visible rank four per hidden spin. The smallest weight is near 5.63×10^{-21}, and no threshold discards a line.

Lines invisible to them. Combine a local fermion removal with a neighboring hidden-spin flip,

$$C_x = \Big(\sum_{y\sim x} S_y^+\Big) c_{x\downarrow}, \qquad S_y^+ = c_{y\uparrow}^\dagger c_{y\downarrow}, \qquad \|C_x\| \le 3.$$

On the singlet reference the output has total hidden spin 3/2 and eta 1/2. The linear probes have hidden spin 1/2 and K commutes with hidden spin, so the complete dynamical cyclic spaces of the two responses are orthogonal. Exact integer residuals certify all 138 spectral lines in the composite probe's reducing block, each with strictly positive weight, and direct calculations of $C^\dagger C$ and $C^\dagger[K, C]$ on the unreduced source independently enclose the total weight and energy moment.

Table 2: Certified charged lines of the centered cube, in source-clock energy units. The first six of each spectrum are shown; the complete tables retain every line and weight.

Line	Linear probe	Relative to first	Composite probe	Relative to first
0	0.361682711806	1.000000000	0.402468515770	1.000000000
1	0.402293044056	1.112281652	0.422089756328	1.048752237
2	0.415174263333	1.147896346	0.438990742922	1.090745551
3	0.438196496809	1.211549467	0.479938217500	1.192486365
4	0.462192207579	1.277894111	0.485785522474	1.207014967
5	0.474429777956	1.311729210	0.494035650369	1.227513783
Largest	3.886434754951		2.974756508168	
Conditional mean	0.49955163275		0.52537938729	

Each composite residue matrix is twice its local weight times one of the source's two kinetic-band projectors, of exact rank four, with 69 lines on each band. Spin and eta ladders give 2,208 visible composite states per charge sector. Their orthogonal union with the linear probes covers 4,032 of the 11,440 states of that number sector, and the block therefore does not exhaust every charged composite. Independent source equations and all 192 spatial energy moments retain the contact contribution and the kinetic/contact interference. The finite spatial projectors supply no continuous momentum dispersion.

The composite response can be read. A compiled pulse instrument implements the success Kraus operator $C_0/3$ in 1,008 source pulses on the authenticated hidden-frame ground preparation, retaining all 7,776 raw selector records, native aborts and coherent field-reference amplitudes. Its success probability is approximately 0.08261623248916, one ninth of the certified composite spectral weight. It detects the total response; no clock resolving the 138 energies is executed.

These lines give one clean exclusion. The visible energy spread is below eleven, so no three distinct lines with one positive unit reproduce the charged-lepton mass ratios. That excludes this finite gap dictionary, and it leaves other sources, refinements and unseen composites untouched.

3.3 Complete conserved quadratics

The labels of these spectra come from a complete classification.

Theorem 3.1 (Complete quadratic conservation)**.** *On a finite connected open rectangular box with V sites and two fermion modes per site, let*

$$H_\mu = \sum_{\langle x, y=x+e_j\rangle} c_x^\dagger (\tfrac{ic}{2}\sigma_j) c_y + \text{h.c.} + U\sum_x n_{x\uparrow} n_{x\downarrow} - \mu N, \qquad c, U > 0.$$

Modulo scalars, the real space of self-adjoint even conserved operators of degree at most two is spanned by $N - V$ and S_1, S_2, S_3 when $\mu \neq U/2$, and by these together with η_x, η_y when $\mu = U/2$, where

$$S_a = \tfrac{1}{2} \sum_x (-1)^{\sum_{j \neq a} x_j} c_x^\dagger \sigma_a c_x, \qquad \eta_+ = \sum_x (-1)^{x_1+x_2+x_3} c_{x\uparrow}^\dagger c_{x\downarrow}^\dagger.$$

The traceless conserved Lie algebra is hidden $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ for H_0 and hidden $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ for the centered generator $K = H_{U/2}$.

Proof. Commutators with H_μ preserve the number change of each term, so the number-preserving and anomalous parts separate. The one-particle restriction gives $[h, q] = 0$ for a one-body matrix q, hence kinetic commutation on the whole Fock algebra and commutation with the doublon count. An input with one spectator and its doublon output isolate each intersite coefficient and force it to vanish. The kinetic condition $q_y = \sigma_j q_x \sigma_j$ then transports one Hermitian root matrix over the connected box, which gives the four stated coefficients. For a pair creator, comparing the vacuum-to-pair entry with its spectator entry gives $U\Delta_{pq} = 0$ on distinct sites. Onsite terms obey $(U - 2\mu) f_x = 0$, and at $\mu = U/2$ the kinetic condition forces the alternating profile, whose real and imaginary parts are η_x and η_y [11]. □

Exact full-CAR coefficient ranks and independent occupation-matrix checks verify the finite instances, and centering changes the conserved pairing algebra as well as the charged energies. Higher-degree currents, Lorentz stress and an interacting chiral continuum are outside the theorem. The radial defect source has its own classification: every conserved sum of onsite even operators is eta-pairing SU(2), and on the mass-64 source every center Pauli current has leakage norm above 1/64 outside the low quartet. The complete native color, weak and hypercharge action therefore cannot be inherited from that onsite-even algebra.

3.4 One-body massless and massive sectors

The same source has exact one-body sectors. On the periodic cube of eight sites per axis a complete spectral detector resolves all 1,024 one-particle modes, including sixteen zeros and six pairs of signed shells, with every pointer and abort outcome retained. A stationary one-particle state cannot occupy only six sites, and the infinite lattice has no square-summable stationary one-particle state.

With centered clock $H - (U/2)N$ and $c = v/a$, the continuum is the free massless field in all eight Weyl valleys, $E = v|\mathbf{p}|$, throughout $0 \leq U$, $U = o(c^{2/7})$ [6]. Supplied quadratic mass coefficients give two Dirac copies at each rest energy $|m_0 \pm |\mathbf{m}||$, with continuum dispersion $E^2 = v^2|\mathbf{p}|^2 + M^2$ and a proved finite-spacing error. This uniform family has at most two distinct positive rest energies, so a direct assignment cannot reproduce three distinct charged-lepton masses. The ordered-spin continuum has nonzero positive-energy bands at every small momentum, and its sharp rest mass and dispersion are undetermined. Matching a light speed fixes length unit $= cT$; determining T requires an independently identified same-generator clock.

4 The charged mass operator

4.1 Pixel capacity and the mass law

The pixel variable is the dimensionless capacity ratio $P = a_{\text{cell}}/\ell_P^2$, distinct from a projector P_j in the memory controller. The declared self-reference relation reads

$$P = \varphi + \sqrt{\pi}\,\alpha, \qquad \varphi = \frac{1 + \sqrt{5}}{2}.$$

In the heat/running construction the inner response supplying α depends on P through a unified scale, a heat-sum equation, low-energy couplings and charged masses, so the loop has the form

$$P \longrightarrow (\alpha_U, v, \text{field census}) \longrightarrow (M_f, J_f) \longrightarrow A_{\text{response}} \longrightarrow \varphi + \frac{\sqrt{\pi}}{A_{\text{response}}}.$$

The same masses and current normalization appear on both sides. A fixed point of a declared response proves consistency of that map, and its identification with the physical electromagnetic response is a further claim.

The controller is a native family memory. It stores one of twelve port addresses, each with a rank-one family projector P_j. A C event rotates the port label, a joint event then applies the rotation of E, and other records reset the address to port two. The fair-coin rates of the C and E events are $p = 1/12$ and $q = 1/6$, and all 128 raw record labels are retained. Declared current probes lower these rates to the values that follow.

The mass law used here is a chosen response architecture with icosahedral registers and determinant normalization. Its charged shape uses a three-phase Koide-type parametrization [7, 8]. The architecture has target-informed provenance: its numerical comparisons are conditional postdictions even when its execution reads no measured masses. The quark exponents, scalar quartic, neutrino operator and physical flavor mixing are not determined from the premises A1–A3 of the registry [25]. Independent validation requires consequences not used to choose it. The operator realization below applies to the displayed $\beta = 4$ branch and to the normalized coefficients of the explicit $\beta = 7$ branch.

4.2 A pixel mass law in native memory dynamics

A chosen simulator specification can realize the conditional pixel-dependent charged mass law on the native gauge-family carrier. The construction uses the twelve-port reset controller and the complete native current instruments. Its design inputs are independent left and right probe durations, a rule enabling feedback on a successful C event, and a common feedback amplitude. The conditional mass-law architecture is target-informed; evaluating that architecture without a charged-mass input does not establish a prospective prediction.

The selected response architecture gives

$$\kappa = \frac{\alpha_U/50}{1 + \alpha_U/31 + \alpha_U^2/310}, \qquad \chi = \frac{\alpha_U^2/512}{1 - \alpha_U/77}, \tag{2}$$

$$\zeta = \frac{(P - \varphi)^2/(21\pi)}{1 - \alpha_U/27 - \alpha_U^2/135}, \qquad r_j = 1 + \sqrt{2}e^{-\chi}\cos\left(\frac{2}{9} + \zeta + \frac{2\pi j}{3}\right), \tag{3}$$

where $\varphi = (1 + \sqrt{5})/2$ and $j = 0, 1, 2$. Its charged coefficients, ordered increasingly, are

$$t_i = \frac{m_i}{v} = \frac{e^{\kappa} r_i^2}{\left(2\,6^{14}\prod_{j=0}^{2} r_j^2\right)^{1/3}}, \qquad \frac{v}{E_*} = P^{-1/2}\exp\left(-\frac{2\pi}{4\alpha_U}\right). \tag{4}$$

The register dimensions and determinant exponent in these formulas are part of the selected architecture. The scale E_* is a source coordinate; assigning it an energy in laboratory units requires a clock calibration.

Let $t_i > 0$ be these three charged coefficients and $\lambda_i = t_i/\sum_j t_j$. For the native projectors P_2, P_3, P_{10}, direct multiplication gives $\operatorname{tr}(P_i P_j) = 1/5$ for distinct indices and the spanning-vector

Gram determinant $d = 2/5 - 2\sqrt{5}/25$. Consequently the matrix $Y = uP_2 + v_1P_3 + wP_{10}$ has elementary eigenvalue invariants

$$\operatorname{tr} Y = u + v_1 + w, \qquad e_2(Y) = \frac{4}{5}(uv_1 + uw + v_1w), \qquad \det Y = duv_1w. \tag{5}$$

Choose its three positive weights as the roots of

$$z^3 - z^2 + \frac{5}{4}e_2(\lambda)z - \frac{e_3(\lambda)}{d} = 0, \qquad w < u < v_1. \tag{6}$$

It follows that $\operatorname{spec} Y = \{\lambda_i\}$.

The complete C-enabled memory coefficient, with $a = p(1-q)$, $b = pq$ and $D = 1 - a^2 + ab$, is

$$Z_C = \frac{(1-a)P_2 + a(1-a)P_3 + abP_{10}}{D} - (1-p)P_2. \tag{7}$$

It has trace p; disabled records have not been removed or normalized away. Put $A = v_1(v_1 + w)$ and $B = v_1(1 - v_1 + w) + (v_1 + w)w$. The inverse equations

$$-Ap^2 + Bp - w = 0, \qquad q = 1 - \frac{v_1}{1 - p(v_1 + w)} \tag{8}$$

give the exact identity $Z_C = pY$. This follows by substituting the second equation and reducing each of the three coefficient differences by the quadratic in the first equation.

At each of the supplied forward and measured-endpoint pixel coordinates, rational Taylor enclosures and polynomial endpoint signs give exactly three roots of (6), and exactly two solutions of (8) satisfying $0 < p < 1/12$, $0 < q < 1/6$. For the forward coordinate the first solution is approximately $p = 0.01915526454$, $q = 0.05474497127$. The native charged-lepton support Casimirs are $9/8$ and 3, so the complete probes implement these rates with the declared durations

$$\theta_L = \frac{\arcsin\sqrt{12p}}{\sqrt{9/8}}, \qquad \theta_R = \frac{\arcsin\sqrt{6q}}{\sqrt{3}}. \tag{9}$$

Both probes retain all thirteen outcomes. Their product retains all 169×128 current and repair labels per old memory input. The covariant binary grouping retains 512 labels per input and all twelve old inputs.

Choose $g = \sum_i t_i/p$. Then gZ_C has exactly the three coefficients t_i. Its diagonalizing unitary acts on the gauge-trivial family factor, commutes with every native gauge current, and preserves the canonical kinetic form. Thus this is an operator realization of the conditional mass law, up to the allowed family frame. It does not assert an additional alternating-group intertwiner. The source-coordinate enclosures also give $\sum_i t_i < 1$ by bounding the cube of that sum.

The coefficient is the generator of a controlled limit of the full correlated memory. Apply $\exp(-ihgX \otimes P_{\text{new}})$ on enabled records and identity on disabled records. Let Ψ_h be this complete classical/quantum channel. Its classical transition satisfies $T = (1-p)R + pS$, where R resets to port two and S is stochastic. For every normalized initial classical/quantum state z_0, including an arbitrary untouched finite reference, the reduced state after N steps of size $h = \tau/N$ obeys

$$\left\| \sum_{\text{memory}} \Psi_{\tau/N}^N(z_0) - \mathcal{U}_\tau\left(\sum_{\text{memory}} z_0\right) \right\|_1 \le \frac{4g\tau}{N(1-p)} + \left(\frac{8}{1-p} + 4\right)\frac{g^2\tau^2}{N}, \tag{10}$$

where $\mathcal{U}_\tau$ is generated by $gX \otimes Z_C$. For a stationary product preparation the first term vanishes.

To prove the bound, let $\Pi z = \pi \sum z$ and $e_k = (I - \Pi) z_k$. The sum of trace norms on memory blocks gives $\|Te\|_1 \le p\|e\|_1$ on zero-sum blocks and $\|\Psi_h - T\| \le 2gh$. Hence $\|e_k\|_1 \le 2p^k + 4gh/(1-p)$. The reduced stationary step and the mean unitary step have the same first derivative; their two Taylor remainders sum to at most $4g^2h^2$. The reduced error due to e_k is at most $2gh\|e_k\|_1$. Summing these errors proves (10).

This limit is in dimensionless simulator time. A finite unread channel need not equal its limiting unitary. The feedback pulse service is an explicit design choice; compilation into a smaller native gate alphabet is a separate requirement. The forward pixel coordinate has no complete physical endpoint proof, and the measured-endpoint coordinate inherits measured-alpha calibration. No physical pole-mass identification, renormalization map, operational energy scale, or complete quark and neutrino spectrum follows from this finite realization.

4.3 Memory correlations beyond the mean unitary

The native stationary memory has three distinct projector rays, because $P_9 = P_{10}$. Summing their complete temporal correlations gives a controlled correction to the mean evolution. Let T be the reduced column-stochastic transition, E its enabled-pulse part, π its stationary column, $\Pi = \pi \mathbf{1}^T$ and $w = E\pi$. The reset identity contracts zero-sum memory vectors by $p < 1$, so the group inverse $\mathcal{R} = (I - T + \Pi)^{-1} - \Pi$ sums every lag. The one-sided covariance and its positive symmetric part are

$$K = \tfrac{1}{2}(\operatorname{diag} w - ww^T) + E\mathcal{R} \operatorname{diag} w, \qquad D = K + K^T \succeq 0,$$

where positivity follows from the limiting covariance of accumulated pulse counts. For $H_i = gX \otimes P_i$ and $L_i = -i[H_i, \cdot]$, the reduced generator is $G_0 + hG_1$, with $G_0 = \sum_i w_i L_i$, $G_1 = \sum_{ij} K_{ij} L_i L_j$. Its antisymmetric coefficient gives the coherent ordering potential $H_{\mathrm{ord}} = -(i/2) \sum_{ij} K_{ij} [H_i, H_j]$; the symmetric part gives the dissipator $-\frac{1}{2} \sum_{ij} D_{ij} [H_i, [H_j, \cdot]]$. Both belong to the same correction, and $H_{\mathrm{ord}} = I_2 \otimes A$ is a family potential with a different auxiliary grading from a scalar mass.

For stationary product preparation, any untouched reference, $h = t/N$ and fixed t, the corrected completely positive semigroup has an explicit rational $O(h^2)$ finite-time diamond-norm bound on the unread channel. Complete record tapes and correlated initial memory need their own terms.

4.4 Complete records, capacity and recovery

The native twelve-address family memory admits an exact smaller encoding. Every raw branch preserves the recurrent set $S = \{2, 3, 9, 10\}$. Let E_S embed four memory addresses into these source addresses. For all 128 raw labels, all twelve original old-memory labels and every real pulse size, the corresponding reduced branch B_y obeys

$$K_y (E_S \otimes I) = (E_S \otimes I) B_y.$$

Multiplication proves equality of all joint records and terminal work states for any initially encoded state, including entanglement with an arbitrary reference. Reset to address 2 and stationary memory qualify. Address 0 admits one fully retained startup event, landing at 2 or 9, followed by exact recurrent encoding. These maps encode one complete law, and physical independence of capacity would need more.

The native matching code obeys the ideal recovery identity $R_y K_y E = \sqrt{w_y} E$, with $\sum_y w_y = 1$, for every retained branch, on both matching codes, both repair laws and all thirty seams. Specified logical evolution between corrected repairs is therefore preserved even conditional on the records, and faithful recovery cannot shift its poles.

Changing feedback duty changes the evolution. The native always-duty and C-duty generators have ratios approximately $1 : 195.117 : 3579.826$ and $1 : 53.588 : 271.403$. Their normalized quadratic traces differ exactly, so neither a family relabeling nor a common clock rescales one into the other. Dividing by event probability changes only the common scale. The specified nine-transition preparation and two-clock inverse have stationary-kernel error at most $1/2579890176$, which controls a first-order kernel approximation.

4.5 Flavor from current-gated memory

Reading a specified memory's angle, reset rule, duty policy and feedback law determines its stationary coefficients, and these are source-derived for that specification. The declared native current gates the complete repair record differently in the up, down and charged-lepton sectors, and the static mean-action service has certified full-action spectra and nontrivial real quark mixing. For $\theta = \pi/5$, duty "always" and unit Higgs norm, the dimensionless coefficients are as follows.

Channel	Three ordered coefficients		
Up	0.00015764215	0.04279127122	0.95705108663
Down	0.00014014156	0.04369077654	0.95616908190
Charged lepton	0.00003179436	0.02148604675	0.97848215888

Every coefficient in the retained real-projector class has $\mathrm{Tr}[H_u, H_d]^3 = 0$, and complex family basis changes cannot create charged-current CP violation. All four default duty policies fail direct identification of the charged-lepton ratios: under a common positive normalization and relative corrections bounded by δ, the unconditional and C-duty cases require $\delta \geq 0.79167$ and $\delta \geq 0.59931$, and E and CE duties have an exact zero coefficient. Other controllers and source angles lie outside this comparison.

4.6 Pixel feedback with one charged-fermion mass prescription

The pixel feedback must use the same mass coefficients as the declared field and memory operators. Let $u = \alpha_U$ and retain

$$M_U/E_* = e^{-2\pi} P^{1/6}, \qquad v/E_* = P^{-1/2} e^{-2\pi/(\beta u)}, \qquad \alpha_i^{-1} = u^{-1} + \frac{b_i}{2\pi} \log(M_U/m_Z).$$

Here E_* is a declared energy unit. The same conditional three-lepton coefficients are inserted into the charged-fermion integral, together with all six declared quark coefficients, including the finite top contribution. With the prescribed factor $1 - 3\alpha_3/\pi$ on the quark terms, the resulting inverse response $A(P)$ defines $P = \phi + \sqrt{\pi}/A(P)$. This kernel supplies no charged-vector or charged-scalar terms, confining hadronic spectral measure, threshold matching, or derivation of the native current's physical Thomson limit.

There is an exact obstruction to substituting the minimal Standard Model census while retaining $\beta = 4$. Assume $P \geq 1$, $M_U \geq m_Z$, positive couplings and threshold-free one-loop coefficients $b = (41/10, -19/6, -7)$. Set $x = \alpha_2^{-1}$ and $y = \alpha_3^{-1}$. Eliminating the unified coupling and scale ratio gives $0 < y \leq x$ and

$$\frac{\pi(18x + 5y)}{46} = 2\pi - \frac{2}{3} \log P + \frac{1}{2} \log\left[\pi \left(\frac{1}{x} + \frac{69}{333x - 218y}\right)\right].$$

The logarithm's argument is at most $8\pi/(5x)$. The function $9\pi x/23 - 2\pi - \frac{1}{2}\log[8\pi/(5x)]$ is increasing and exceeds one at $x = 6$, so $x < 6$ and both nonabelian couplings exceed $1/6$. Their heat-sum denominators are at least one; direct outward summation with infinite-tail bounds places their combined numerators below $1/80$. The required pixel entropy is $P/4 \geq 1/4$, a contradiction. This excludes the stated scale and census combination, independently of the readout kernel.

Two other declared maps have certified local fixed points. The supersymmetric running proxy with $\beta = 4$ gives $P \simeq 1.630971514159316$ and $A \simeq 137.000994766342$. The minimal census with the explicit scale choice $\beta = |b_3| = 7$ gives $P \simeq 1.628533369611833$ and $A \simeq 168.815082928453$. The first requires extra supersymmetric fields absent from the native quadratic carrier. The second supplies a scale choice, without identifying the Higgs scale physically with color transmutation. For each map, verified implicit-root signs, infinite heat tails and interval automatic differentiation give a strict self-map and $|g'| < 0.00734$ on the reported coupling interval. Uniqueness is local to that interval.

Rational interval inversion over both complete root enclosures admits both native memory solutions, with $0 < p < 1/12$ and $0 < q < 1/6$. The gain $g = \sum_j (m_j/v)/p$ preserves the unconditional duty probability, so the same full-record memory coefficient has the required three singular energies. Independent simultaneous-root and quadrature checks accompany the proof. These are conditional consistent feedback calculations; neither is a complete physical particle or fine-structure prediction.

The beta coefficients in the feedback calculation are obtained from the actual internal-current traces with the intended chiral field multiplicities. This assumes that those are the surviving continuum fields, and it does not count or remove spatial regulator valleys or doublers. The minimal-census root and the supersymmetric proxy are different scale designs, not two calibrations of one proved physical theory. The absence of a positive solution for the minimal census with $\beta = 4$ is a mathematical inconsistency of that specified combination; it is not an exclusion of other admitted OPH designs.

The three increasing coefficients returned by these closed conditional maps are dimensionless singular coefficients m_i/v.

Scale design	m_1/v	m_2/v	m_3/v
Supersymmetric proxy, $\beta = 4$	2.07203042×10^{-6}	4.28430204×10^{-4}	7.20520710×10^{-3}
Minimal census, $\beta = 7$	2.07103435×10^{-6}	4.28253912×10^{-4}	7.20232110×10^{-3}

The interval certificate encloses these values and both admissible memory inverses over each complete root box, so the table and the controller refer to the same conditional source coordinates.

5 Localizing the same current and mass operators

5.1 Topology, total energy and the radial source

The seed of the defect hypothesis is a precise finite frustration statement. For a finite connected graph, abelian edge offsets b_e and positive weights w_e, define

$$\Phi_b(x) = \sum_{e:u\to v} w_e \, \mathbf{1}[x_v - x_u \neq b_e].$$

If a cycle has nonzero signed offset sum, then $\min_x \Phi_b(x) > 0$: the possible costs form a finite set of positive-weight subset sums, and a zero cost would solve every edge equation and force zero cycle sums [3]. This is frustration at fixed compatibility data. Protection of b against an update that changes its class, localization of its energy and a rest mass all need more.

For a one-cochain b on the full carrier complex, face curvature is $F = \delta b$ and $b(\partial c) = F(c)$ for every two-chain c. The icosahedral boundary has trivial first homology, and the native load difference $b_{uv} = n_v - n_u$ is exact, so $F = 0$ and all loop sums vanish. An actual defect invariant therefore requires additional transport or field data.

Consider the two-spin cubic hopping operator with forward link $i\sigma_j/2$ and its Hermitian reverse. A declared radial onsite texture is

$$B_x = M \sum_{j=1}^{3} (-1)^{x_j} \frac{x_j}{|x|} \sigma_j, \qquad B_0 = 0.$$

On the odd radius-one cube at $M = 64$, this literal source has exactly two zero modes and a complementary absolute-energy gap at least 62. The normalized kernel frame W has scalar local density $W_x^\dagger W_x = \rho_x I_2$, with $\sum_x \rho_x = 1$. All 28 retained winding matrices have exact spectral enclosures with every multiplicity and source-matrix rounding error controlled. For every retained odd-degree cube, inversion and complete spectral exhaustion prove exactly two zero modes, including weak-mass cases. Degree-three six-level windows hold two zeros and four nonzero levels, which is no count of generations. A complete-Fock contact theorem admits 34 of 84 supplied contact cases, with a ground pair and unique neighboring charged states.

A radial mass profile also needs a total energy law. In the selected ordered-spin rotation orbit, each admissible charged shell costs at least $\pi J \kappa |q|$ with $\kappa > 0$. A hedgehog retaining nonzero degree on arbitrarily many disjoint shells therefore has infinite excess energy in that ungauged identification, and a fermion bound-mode gap omits this background cost. Screening, compact opposite-charge textures or another specified finite-energy structure are substantive dynamical choices. The compact pair below provides the second kind.

5.2 One local carrier for defect states, currents and mass coefficients

The radial defect operator admits a direct realization of the native matter currents and a prescribed family mass operator on the same localized modes. The construction is a choice of simulator carrier and coherent pairing service. Its finite operator identity is exact; the choice is not claimed to follow uniquely from overlap consistency.

Let h be the radial operator on $\ell^2(\Lambda) \otimes \mathbb{C}^2$, with its normalized kernel frame $W : \mathbb{C}^2 \to \ker h$. Use the strong texture domain, in which the kernel has dimension two and the complementary spectrum has absolute energy at least $d > 0$. With

$$J = i\sigma_2, \qquad D_x = (-1)^{x_1+x_2+x_3}, \qquad C = DJ,$$

the source identities are

$$C^T = -C, \quad C^\dagger C = I, \quad hC = -C\overline{h}, \qquad hW = 0, \quad W^\dagger W = I_2, \quad C\overline{W} = WJ.$$

The last identity fixes the quaternion frame through evaluation at the origin and its scalar normalization Gram matrix.

Choose the canonical anticommutation relation algebra on $\ell^2(\Lambda) \otimes \mathbb{C}^2 \otimes M_{15} \otimes \mathbb{C}^3$. Here M_{15} is the native matter representation, transported by its actual source intertwiner; the three family copies are declared inputs. For a symmetric all-left Higgs kernel $K \in M_{45}(\mathbb{C})$, define

$$\Delta = C \otimes \overline{K}, \qquad B = \begin{pmatrix} h \otimes I & \Delta \\ \Delta^\dagger & -\overline{h} \otimes I \end{pmatrix}, \qquad E = \operatorname{diag}(W \otimes I, \overline{W} \otimes I).$$

The antisymmetry of Δ makes it a valid local creation pairing. Multiplication gives the exact isometric intertwiner

$$BE = EB_0, \qquad B_0 = \begin{pmatrix} 0 & J \otimes \overline{K} \\ (J \otimes \overline{K})^\dagger & 0 \end{pmatrix}. \tag{11}$$

Every singular value of K therefore appears with both signs and twice the multiplicity in the localized Nambu operator, including zero values. Nambu coordinates duplicate annihilation and creation operators; they introduce no physical mirror carrier.

For every actual anti-Hermitian native current G, the map $W \otimes I$ intertwines $I \otimes G$ with $I_2 \otimes G$. In Nambu coordinates the generator is $\mathrm{diag}(G, \overline{G})$; Hermitian charges use $\mathrm{diag}(Q, -\overline{Q})$. The Higgs background transforms with matter under the uniform internal gauge action. A fixed broken vacuum does not make every gauge generator commute with its mass operator. The resulting map of canonical anticommutation relation algebras intertwines quadratic Heisenberg evolution. On even states, tensoring with the normalized trace on the orthogonal complement gives an exact state inclusion with partial-trace recovery. This complement state is not a vacuum.

Use the positive family matrix $M(P)$ supplied by the memory construction, whose eigenvalues are the retained charged-lepton coefficients $m_i(P)/v(P)$. Choosing $H = (0, v(P)/\sqrt{2})$ and $Y_e = \sqrt{2}M(P)$ realizes those coefficients in $K/v(P)$ itself. The coherent pairing service implements the stationary-mean memory coefficient times its declared gain. Equation (11) does not identify the unread finite memory channel with this Hamiltonian. The calibration provenance of the retained family law is unchanged.

An explicit record service realizes the same Hamiltonian in a controlled weak-step limit. Apply the native current probes to fresh auxiliary legs in their invariant preparations, so their complete twelve-state memory transition is independent of the spatial field. For the updated memory port j choose the field pulse generated by

$$H_r = H_0 + \mathbf{1}_C(r)\, gL(P_j),$$

where H_0 contains the radial kinetic term and the chosen quark pairing, L is the local lepton pairing map, and g is the source gain. Every H_r has the same core intertwiner. Linearity gives stationary mean $\overline{H} = H_0 + L(gZ_C)$, precisely the Hamiltonian above. If $\|H_r\| \le M$ on the fixed finite Fock space, the reset contraction of the complete memory law and a second-order unitary expansion give

$$\left\| \rho_N - e^{-iT\overline{H}} \rho_0 e^{iT\overline{H}} \right\|_1 \le \frac{4MT}{N(1-p)} + \left(\frac{8}{1-p} + 4 \right) \frac{M^2 T^2}{N}.$$

The bound includes arbitrary finite untouched references and initial memory correlations; stationary product memory removes the first term. For the displayed cube, $M = 2430 \cdot 66 + 1215(1+g)$ suffices. This chosen coherent service uses every record. Its finite unread channel need not be unitary, and its bound is not uniform in volume.

For a complete finite witness, choose the retained quark design $Y_u = \mathrm{diag}(6^{-6}, 6^{-3}, 1)$ and $Y_d = \mathrm{diag}(6^{-6}, 6^{-4}, 6^{-2})$, and set the three minimal neutrino coefficients to zero. Their exponents are design inputs, not topological index conclusions. On the radius-one cube there are 27 sites, 2430 physical annihilator modes and 4860 Nambu rows. The localized sector has 90 annihilators and 180 Nambu rows: 84 positive, 84 negative and 12 zero rows. These numbers count operator degrees of freedom, not observed species. In particular the zero rows do not establish a relativistic massless neutrino sector.

The full complement is retained. Indeed, with $T = \mathrm{diag}(I, C^\dagger \otimes I)$,

$$T^\dagger B T = \begin{pmatrix} h \otimes I & I \otimes \overline{K} \\ I \otimes K & h \otimes I \end{pmatrix}.$$

Its energies are the sums $\lambda(h) \pm s_j(K)$, not a relativistic square-root dispersion. If $\|K\| \le M$ and $2M < d$, the localized band lies in $[-M, M]$ and the complement has absolute energy at least $d - M$. This is single-quasiparticle isolation, not isolation of all many-body energy sums. In the displayed design, choosing $v(P)$ as energy unit, radial hopping one and texture coefficient 64 admits $d = 62$ and $M = 1$.

The common spatial density follows from $W_x^\dagger W_x = \rho_x I_2$; inversion gives mean position zero. A Fourier transform supplies a momentum distribution, without making the localized state a momentum eigenstate. Occupation of these modes does not create the supplied texture winding. Original site-resolved repair can leak out of the kernel, and interactions can mix the two factors. Thus the exact common finite operator supplies neither defect formation, physical rest masses nor an interacting relativistic continuum.

The certificate retains all twelve native current operators, and its complete finite controls include an unstaggered-pairing counterexample.

5.3 Compact opposite-degree textures and their isolated modes

A finite-energy texture can carry opposite local winding numbers while approaching a constant exterior. On the two-spin cubic hopping carrier, choose an integer $a \ge 1$, write $R = a + 1$, and define

$$f(x, y, z) = \begin{cases} (x, y, z^2 - a^2), & \max(|x|, |y|, |z|) \le R, \\ (0, 0, 1), & \text{otherwise.} \end{cases}$$

Set $n = f/|f|$ away from $c_\pm = (0, 0, \pm a)$ and $n(c_\pm) = 0$. The Hamiltonian is $h = T + Mn \cdot \sigma$, where the nearest-neighbor hopping has $\|T\| \le \sqrt{3} < 2$ and $M \ge 8$. The texture and its amplitude zeros are explicit design inputs. The complete unit cubic shells about c_+ and c_- have degrees $+1$ and -1 under their normalized barycentric interpolation; the exterior shell is constant. Exact rational regular-value counts check every oriented triangle. This degree classifies the interpolated texture and is not a fermionic superselection charge.

For the selected ordered-spin reference, fill the two core directions by the exterior direction. Each rotated-ground source bond has excess energy in $[0, J]$. With $s = 2a + 3$, precisely $3s^2(s + 1)$ bonds touch the support cube, so

$$0 \le \Delta E_{\text{spin}} \le 3s^2(s + 1)J.$$

The two zero fermion masses and the spin-core fill are distinct declared parts of the construction. The second-quantized Hamiltonian differs from that of the uniform texture by an operator of norm at most $2M(s^3 - 1)$. Its vacuum-energy difference has the same bound, independently of exterior volume. These are total texture-energy bounds; they do not assign a constituent rest mass to a localized mode.

Projection onto the two core sites gives $h = \left(\begin{smallmatrix} 0 & B^\dagger \\ B & D \end{smallmatrix}\right)$. The exterior block has no spectrum in $(-M + 2, M - 2)$, and $\|B\|^2 = b^2$, where $b^2 = 7/4$ for $a = 1$ and $b^2 = 3/2$ otherwise. There are exactly four low eigenvalues. Their Feshbach equations give

$$|E| \le r := \frac{b^2}{M - 4}, \qquad \frac{\|\psi_{\text{exterior}}\|^2}{\|\psi_{\text{cores}}\|^2} \le \frac{b^2}{(M - 2 - r)^2}.$$

These bounds are uniform in volume. For a cube of radius $L \ge R + 1$, resolvent path expansion and monotone Feshbach eigenvalue crossings give

$$|E_j(L) - E_j(\infty)| \le \frac{2b^2}{M - r - 2} \left(\frac{2}{M - r}\right)^{2(L+1-R)}.$$

Thus the infinite lattice also has precisely four isolated modes in $[-2,2]$. At $a=1$, $M=64$, $L=3$, complete rational interval certification of all 686 eigenvalues excludes zero from all four low levels and from their infinite-volume limits. The numerical levels are approximately ± 0.003905899328 and ± 0.003906437151. The finite-volume filled-sea energy difference is approximately 127.9700 in the same hopping units.

Let W span this four-dimensional band, $A = W^\dagger h W$, and $C = D_{\mathrm{stagger}} i\sigma_2$. Particle-hole symmetry gives $C\overline{W} = WS$ with $S = W^\dagger C\overline{W}$ antisymmetric and unitary. The actual twelve native current operators intertwine through $W \otimes I_{45}$. The same local pairing $\Delta = C \otimes \overline{K}$ has the exact core

$$\begin{pmatrix} A \otimes I & S \otimes \overline{K} \\ (S \otimes \overline{K})^\dagger & -\overline{A} \otimes I \end{pmatrix}.$$

Its energies are those of A plus or minus the singular values of K. Consequently the compact texture shifts the retained family coefficients; dropping A gives an incorrect operator inclusion. A local Pauli perturbation at the two cores preserves every shell datum while changing these energies. Winding alone therefore does not pin a zero mode for this carrier. The construction supplies a compact background and controlled localized spectrum, without autonomous formation, stability, a relativistic mass shell or physical particle identification.

5.4 Protected texture records and controlled band transport

A declared local repair law protects the compact pair's two winding charges. Let f_x be its unnormalized reference texture and fix the zero cores and constant exterior. At every noncore site retain a rational vector v_x with $\|v_x - f_x\| \le \eta \|f_x\|$, $0 < \eta < 1/4$, using $v_x/\|v_x\|$ in the actual Pauli potential. A one-vertex proposal is accepted only if it satisfies this budget and has nonnegative scalar product with the old vector and every other vertex of every incident shell triangle. Shared-face data are checked on both shells. Rejected proposals remain recorded identity operations.

Proposition 5.1 (Guarded winding and uniform isolated band)**.** *Every admitted finite history preserves the two shell degrees. Its spatial Hamiltonian has exactly four isolated modes with $|E| \le r = b^2/(M-4)$ and complementary spectrum outside $(-g,g)$, $g = M-2$, where $b^2 = 7/4$ for adjacent unit shells and $b^2 = 3/2$ for separated shells. For two admitted band projectors Π_v, Π_w,*

$$\|\Pi_w - \Pi_v\| \le \beta = \frac{4M\eta c}{d^2}, \qquad c = \frac{g+r}{2}, \quad d = \frac{g-r}{2}.$$

The statements are uniform in the exterior volume.

Proof. During an accepted replacement, every interpolated shell vector is a convex combination of nonzero vectors with pairwise nonnegative scalar products. Its squared norm is at least $\sum_i t_i^2 \|v_i\|^2 > 0$, so normalization gives a shell homotopy. Every noncore onsite eigenvalue stays $\pm M$ and the four core coordinates stay zero; the compact-source Schur proof therefore gives the same band count and localization bounds. Normalization gives $\|H_w - H_v\| \le 4M\eta$. On $|z| = c$ both resolvents have norm at most $1/d$. Their identity and the contour formula for the two projectors give the displayed bound. □

For $\beta < 1$, define

$$S = \Pi_w \Pi_v + (1-\Pi_w)(1-\Pi_v), \qquad U = S(S^*S)^{-1/2}, \qquad S^*S = 1 - (\Pi_w - \Pi_v)^2.$$

Since $S\Pi_v = \Pi_w S$ and the positive factor commutes with Π_v, U is unitary and $U\Pi_v U^* = \Pi_w$. The literal particle-hole relation $H_v C = -C\overline{H_v}$ implies $\Pi_v C = C\overline{\Pi_v}$, and the polar formula gives

$UC = C\overline{U}$, hence $UCU^T = C$. Thus $U \otimes I_{45}$ transports all twelve uniform native currents and the same pairing tensor $C \otimes \overline{K}$. For $W' = UW$, the pairing matrix $W'^* C \overline{W'}$ is unchanged, while $A_w = W'^* H_w W'$ retains the actual changed spatial energies. Second quantization also transports $N_v = d\Gamma(\Pi_v \otimes I_{45})$ and $\lambda(N_v - 90)^2$.

The explicit source uses $M = 64$, $\eta = 1/512$ and four equiprobable proposals at each record: two opposite transverse changes at the shared-face center, one forbidden zero and one forbidden core change. Its three texture states have transition rows $(1/2, 1/4, 1/4)$, $(0, 3/4, 1/4)$ and $(0, 1/4, 3/4)$. All twelve state/proposal records are retained. Each branch applies its texture transition and $\sqrt{q_r}\,\Gamma(U_r)$, with identity control for a repeated texture, so the complete instrument preserves trace. The band is transported exactly at record endpoints. Omitting U_r instead gives leakage probability at most $\min(1, n^2\beta^2)$ after n quenches and intervening source evolution; the numerical controls exhibit nonzero leakage.

A nearest-neighbor snake through the $N = 2(2L+1)^3$ spatial-spin coordinates compiles each U by adjacent Givens rotations: at most $N(N-1)/2$ two-mode gates and N onsite phases. The complete $N = 686$ controls replay 234,955 two-mode gates per nontrivial texture pair. Addressed quadratic pulses, their timing and drift cancellation are supplied services; this is not a compilation into unchanged C/E repair. Intermediate pulses need not preserve the band, and the controller supplies the endpoint energy change, bounded by $2r$ on the one-body band. Erasing the guards permits zero-degree textures with the same four-mode count. The result establishes controlled protection and transport on this law, without autonomous core motion, local Gauss dressing, physical species identification or relativistic dispersion.

5.5 Gauge-covariant paired compact bands

The compact source extends to supplied classical unitary internal links on its 343-site radius-three lattice. The gauge paper [5] proves that with onsite texture amplitude 64 and pairing of norm at most one, the complete Bogoliubov–de Gennes operator keeps a separated band whose particle grading defines 180 canonical annihilation modes and an integer occupation N_r. The quartic $\lambda(N_r - 90)^2$ of Section 5.6 is therefore well-defined on every link background and equals the source compact interaction at trivial links. Transforming link and pairing together preserves band and interaction covariantly, and declared polar controllers transport the grading between records.

The same paper gives the separate native quantum-link design: a 60-state link on the five matter blocks with exact local non-Abelian Gauss constraints on the full Fock space, and a conditional rotor completion that leaves five link-energy differences unselected. A literal native two-site raising-link example has a noninteger naive occupation even on a Gauss singlet, so the canonical band above does not transfer to operator-valued links by substitution. A dynamical quantum-link realization of the interacting compact band is not established here.

5.6 A quartic interaction on the complete compact band

The spatial source is the cubic operator $h = T + Mn \cdot \sigma$, $\|T\| < 2$, $M \geq 8$. Its direction is the normalization of $(x, y, z^2 - a^2)$ inside the cube of radius $a + 1$ and a constant outside, with zero amplitude at the two cores $(0, 0, \pm a)$. The four-band bound is $r = b^2/(M-4)$, where $b^2 = 7/4$ for $a = 1$ and $b^2 = 3/2$ otherwise. The spectral and particle–hole identities hold for this source by the compact-pair construction. Let W contain the four localized modes and set $A = W^\dagger h W$. The isometry $E = W \otimes I_{45}$ embeds 180 annihilators into the complete finite field. The current and pairing intertwiners hold for every complete memory record, with A retained in the quadratic core.

Write $P = EE^\dagger$ and choose

$$V = \lambda(N_W - 90)^2, \qquad N_W = d\Gamma(P), \qquad \lambda \geq 0. \tag{12}$$

This is an explicit controller design. The particle–hole identity $C\overline{W} = WS$ ensures that the pairing has no localized/complementary cross term. All terms are even, so $H_r^{\text{full}} = H_{r,W} \otimes I + I \otimes H_{r,\perp}$ on every record. The interaction is parity even and commutes with all twelve native internal currents. In the localized occupation basis it contains the quartic term $2\lambda \sum_{i<j} n_i n_j$. The pairing Hamiltonian does not conserve N_W, and its commutator with V need not vanish. Every record preserves the same localized algebra because V acts entirely on that factor.

Hold the texture fixed throughout the memory averaging interval. The complete record limit has a bound independent of exterior volume. Writing the compact energies as $\{-e_2, -e_1, e_1, e_2\}$ with $e_i \leq r$, the kinetic norm is $45(e_1 + e_2) \leq 90r$. If g is the source feedback gain, the pairing bound and $0 \leq N_W \leq 180$ give

$$\|H_{r,W}\| \leq M_W = 90r + 90(1 + g) + 8100\lambda.$$

The radial two-column zero-mode construction is a separate special case with ninety annihilators, centre 45 and bound $45(1+g)+2025\lambda$. The actual memory transition has the reset decomposition $T = (1-p)R + pS$, and contracts zero-sum operator-valued memory blocks by at most p. At step size t/n, its reduced core state consequently satisfies

$$\|\rho_{W,n}(t) - e^{-it\bar{H}_W}\rho_W(0)e^{it\bar{H}_W}\|_1 \leq \frac{4M_W t}{n(1-p)} + \left(\frac{8}{1-p} + 4\right)\frac{M_W^2 t^2}{n}. \tag{13}$$

The estimate includes arbitrary initial classical–quantum memory correlations and a finite untouched reference. The first term vanishes for stationary product memory. The mean Hamiltonian contains the same memory-derived mass matrix and the interaction (12). Uniformity concerns this fixed localized algebra under its source isometries; it gives no uniform continuum estimate for all spatial observables.

Equation (13) compares the unread core. There is a separate estimate for the full native history register. For a fixed initial memory label, purify the core/reference state. Write p_ω for the field-independent history probability, $\psi_\omega = U_\omega \psi$, and $\bar{\psi} = e^{-it\bar{H}_W}\psi$. If δ is the unread trace-norm bound, then $\sum_\omega p_\omega(1 - |\langle\bar{\psi}|\psi_\omega\rangle|^2) \leq \delta/2$. The pure-state distance identity and Jensen's inequality bound the full history/core trace distance from $\sum_\omega p_\omega |\omega\rangle\langle\omega| \otimes |\bar{\psi}\rangle\langle\bar{\psi}|$ by $\sqrt{2\delta}$. Partial trace covers mixed inputs. For initial classical memory/core correlations, condition on the initial memory label and retain that tag in the ideal history law; convexity preserves the same bound. The complete final record/core rate is therefore $O(n^{-1/2})$. Opposite qubit phase records saturate this square-root relation. No uniform estimate for rare conditioned histories or intervening field measurements follows.

The interaction is defined through a spectral projector and need not be strictly onsite. For the radial special case, the resolvent estimate at texture coefficient 64 and $W_x^\dagger W_x = \rho_x I_2$ gives, for the projection A_L onto $|x|_\infty < L$,

$$\|(I - A_L)E\|^2 \leq \epsilon_L = \frac{3}{2 \cdot 62^2}\, 32^{-2(L-1)}.$$

Thus $\|P - A_L P A_L\| \leq 2\sqrt{\epsilon_L}$, and finite rank gives $\|N_W - d\Gamma(A_L P A_L)\| \leq 360\sqrt{\epsilon_L}$. The radial interaction $\lambda(N_W - 45)^2$ is exponentially localized about the prescribed origin, without being a strictly onsite interaction.

On this radial subspace there is an exact obstruction for the onsite density alternative. On the two-particle sector let $D_g = \sum_x g_x \bigwedge^2 P_x$ and $E_2 = \bigwedge^2 E$. Direct compression gives

$$E_2^\dagger D_g (I - E_2 E_2^\dagger) D_g E_2 = \left[\sum_x g_x^2 \rho_x^2 - \left(\sum_x g_x \rho_x^2 \right)^2 \right] I. \tag{14}$$

Writing $c = \sum_x \rho_x^2 < 1$, Cauchy–Schwarz bounds the bracket below by $(1-c)\sum_x g_x^2 \rho_x^2$. Every nonzero member of this onsite density class leaks from the source subspace on which it has support. This statement does not exclude a dressed low-energy subspace or other local interaction tensors.

Connected dynamics and spectral shifts. The compact mean pairing operator has 84 charged two-mode blocks and twelve unpaired neutrino modes. Select the two spatial pairs of one charged-lepton singular value m. Fix one occupation in each of the other 82 blocks and fill six neutral modes. Pair creation and annihilation kill these spectators, whose number is 88. Subtracting their fixed energy gives the exact reducing four-mode Hamiltonian

$$H_4 = e_1(n_1 - n_2) + e_2(n_3 - n_4) + m(a_1^\dagger a_2^\dagger + a_2 a_1 + a_3^\dagger a_4^\dagger + a_4 a_3) + \lambda(N_4 - 2)^2.$$

This is an invariant sector of the mean Hamiltonian, not its asserted vacuum or an invariant sector of every finite memory pulse. Pair parities split its sixteen levels into the four even/even values $0, 4\lambda, 2\lambda \pm 2\sqrt{\lambda^2 + m^2}$, four odd/odd values $\pm e_1 \pm e_2$, and eight mixed values $\lambda \pm e_i \pm m$. Consequently

$$E_0 = \min\{2\lambda - 2\sqrt{\lambda^2 + m^2}, -e_1 - e_2, \lambda - m - \max(e_1, e_2)\}.$$

This enumeration is a complete diagonalization by pair parity. For the unique odd/odd ground, a positive-energy orbital has two addition-plus-removal poles $e_1 + \lambda \pm m$, each of residue $1/2$. At $M = 64$, half-separation one and $\lambda = 1/128$, rational coefficient and exterior-volume enclosures place all three inherited charged-lepton slices in this ground sector:

Quadratic coefficient m	Lower pole	Upper pole
0.000002072030404	0.01171632730	0.01172047136
0.0004284302055	0.01128996912	0.01214682953
0.007205207119	0.004513192209	0.01892360645

The unit is the declared source pulse energy. These are slice poles, without a physical mass assignment or a determination of the full interacting vacuum. The compact exterior error δ_L bounds each displayed pole error by δ_L in this certified sector; no complementary mode is discarded.

The full 180-mode ground also has an explicit variational bracket. List the charged singular values, including colour multiplicities, as m_j, $j = 1, \ldots, 21$. Each occurs in two pairing blocks at each spatial energy e_i, with quadratic levels $\{\pm m_j, \pm e_i\}$. The twelve neutral modes contribute $-3(e_1 + e_2)$, giving

$$E_{\mathrm{quad}} = -2 \sum_{j=1}^{21} [\max(m_j, e_1) + \max(m_j, e_2)] - 3(e_1 + e_2).$$

Positivity of V makes this a lower bound. A quadratic product ground has mean occupation 90 and number variance one per even pairing block, zero per odd block. If b blocks are even, its trial

energy is $E_{\mathrm{quad}} + \lambda b$. A completely blocked trial state has exact occupation 90, zero interaction energy and kinetic energy $-45(e_1 + e_2)$. Thus, retaining every localized mode,

$$E_{\mathrm{quad}} \le E_{\mathrm{full}} \le \min\{E_{\mathrm{quad}} + \lambda b, -45(e_1 + e_2)\}.$$

The specified coefficient witness has $b = 28$. These bounds constrain the full vacuum energy without identifying its excitation spectrum.

The radial control sets $e_1 = e_2 = 0$. Its ground energy is $2\lambda - 2\sqrt{\lambda^2 + m^2}$, and its single-mode spectral poles and weights are

$$\omega_\pm = 2\sqrt{\lambda^2 + m^2} - \lambda \pm m, \qquad w_\pm = \frac{1 \mp m/\sqrt{\lambda^2 + m^2}}{2}.$$

In this radial control the cross-pair density cumulant is $-\lambda/(4\sqrt{\lambda^2 + m^2})$, while its normal and anomalous cross-pair two-point contractions vanish. Hence positive λ gives a non-Gaussian connected interaction. These spectral shifts preclude identifying the quadratic mass coefficients with the interacting excitation energies without further calculation.

A common finite energy and modular response. For the same full localized Hamiltonian choose $\rho_\beta = Z^{-1} e^{-\beta \bar{H}_W}$, $\beta > 0$. This state is faithful. A global spectral reset with Kraus operators $\sqrt{p_i}|i\rangle\langle j|$ prepares it and preserves total probability. That reset is a declared service, not an implementation by the original site repair. For $H(s) = \bar{H}_W + sX$, Duhamel differentiation gives

$$\rho'_\beta(0) = -\beta \int_0^1 \rho_\beta^u (X - \langle X\rangle) \rho_\beta^{1-u}\, du.$$

The positive logarithmic-mean kernel yields $F' = \langle X\rangle$, $F'' = -\beta C_{\rho_\beta}(X, X) \le 0$, and $S' = \beta(E' - \langle X\rangle)$, where C_ρ is the Bogoliubov–Kubo–Mori covariance. The noncommuting choice $X = (N_4 - 2)^2$ is checked against an independent matrix-exponential derivative. These are finite identities for one nonquadratic Hamiltonian. They supply no geometric modular flow, local relativistic stress, area normalization or geometric entropy stationarity, and therefore imply no Einstein dynamics. Exact characteristic polynomials, complete finite spectra, record controls and the onsite leakage calculation accompany the proof.

Intervening observations. The bounds above extend to adaptive experiments. On the compact 180-mode core at $\lambda = 1/128$, every record's localized Fock generator obeys $\|H_{r,W}\| \le M_W \le 7869/32$. Between m evolution intervals, apply arbitrary complete quantum instruments to the field and any auxiliary observer register, chosen from every retained native and detector record, with all failure outcomes kept. Replacing actual interval channels by ideal mean-dynamics channels one at a time bounds the complete-history distance by $\sum_j \min\{2, \sqrt{2\delta_j}\}$, where δ_j is the unread bound of interval j. The native memory continues under its actual transition law, since conditioning does not justify resetting it to a stationary product. Exact resource schedules and a complete finite execution check the correspondence. The bounds concern a declared temporal refinement, and adapting at every microscopic step needs its own budget.

5.7 Work on the moving compact interacting band

A deformation must act on the same Hamiltonian used to identify the matter states. On the compact opposite-degree texture, vary the actual hopping by $h_s = sT + MV$, with $1/2 \le s \le 3/2$,

$M \geq 8$ and $\|T\| < 2$. The texture, memory coefficients and quartic coupling are fixed. The four-dimensional spectral band has energies in $[-3, 3]$ and separation at least $M - 6$ from its complete complement. Let P_s be its projector, C the source particle-hole matrix, and K_ω the internal pairing kernel on a complete memory record ω. The same source law is

$$N_s = \mathrm{d}\Gamma(P_s \otimes I_{45}), \qquad H_{\omega,s} = \mathrm{d}\Gamma(h_s \otimes I_{45}) + \mathrm{pair}(\mathsf{C} \otimes \overline{K_\omega}) + \lambda(N_s - 90)^2.$$

Its exact force conjugate to the hopping control is

$$\partial_s H_{\omega,s} = \mathrm{d}\Gamma(T \otimes I_{45}) + \lambda\{N_s - 90, \mathrm{d}\Gamma(P_s' \otimes I_{45})\}. \tag{15}$$

The second term follows by differentiating the noncommuting square. Freezing the localization projector omits this contribution to source work.

Proposition 5.2 (Localized force with the moving band retained)**.** *The band admits a unitary comparison frame W_s in which its pairing form is constant and its intrinsic force is*

$$J_s = \mathrm{d}\Gamma((W_s^* T W_s) \otimes I_{45}), \qquad \|J_s\| \leq 360, \qquad \|P_s'\| \leq \frac{6}{M-6}.$$

For two even localized states with the same even complement reference, their contrast of (15) *equals their contrast of J_s. The bounds are independent of the containing finite cube.*

Proof. For band vectors p and complementary eigenvectors q, $\langle q, P_s' p\rangle = \langle q, Tp\rangle/(E_p - E_q)$ and the diagonal blocks of P_s' vanish. Hence $\|P_s'\|^2 \leq \|P_s'\|_{\mathrm{HS}}^2 \leq 8\|T\|^2/(M-6)^2 < 36/(M-6)^2$. Set $L_s = [P_s', P_s]$ and solve $U_s' = L_s U_s$. Then $P_s = U_s P_1 U_s^*$ and $W_s = U_s W_1$. Particle-hole covariance gives $\mathsf{C}\overline{U_s} = U_s \mathsf{C}$, so the pairing form is fixed. Differentiating $A_s = W_s^* h_s W_s$ gives $A_s' = W_s^*(T + [h_s, L_s])W_s = W_s^* T W_s$: h_s is block diagonal and L_s off diagonal. Number and pairing therefore have no derivative in this transported frame. The 180-mode second-quantization norm is at most $180\|T\| < 360$. In an even product state the cross-band terms have zero expectation; subtracting the same complement reference cancels its remaining energy. □

At fixed s, a complete-memory trace-norm error δ consequently gives a force-readout error at most 360δ. The comparison transporter is an explicit mathematical identification; Hamiltonian evolution with changing s does not implement it automatically. For the supplied Gibbs reference of the same mean Hamiltonian at fixed $\beta > 0$, Duhamel differentiation gives

$$F'_{\mathrm{thermal}} = \mathrm{Tr}(\rho_s J_s), \qquad S' = \beta(E' - \mathrm{Tr}(\rho_s J_s)).$$

The complete compact source and a reducing sixteen-state interacting slice check these derivatives, retaining the spatial splitting and work performed by the hopping control. The hopping amplitude is a declared control. Its work derivative supplies no identification with a metric variation or conserved Lorentz stress, and this finite Gibbs identity supplies no regional geometric modular flow, area normalization or fixed-volume generalized-entropy stationarity.

6 Comparison with the measured spectrum

6.1 Observables and conventions

At finite volume the spectral measure of a prepared operator O is

$$d\mu_O(\omega) = \sum_n |\langle n|O|\Omega\rangle|^2 \delta(\omega - (E_n - E_\Omega))\, d\omega.$$

A visible eigenvalue requires nonzero residue in the actual detector channel, and the physical dictionary must transport O, the state, the Hamiltonian, current charges and the clock together. Localized wavefunctions are not momentum eigenstates, and a Fourier transform of a pinned mode is no moving-defect dispersion experiment.

In a justified relativistic regime a stable massive excitation has a common rest-energy and propagation relation. An unstable particle requires a resonance convention and width [15]. Charged sectors coupled to a massless photon require an infrared treatment rather than an assumed isolated sharp pole [17, 18], and color confinement prevents reading a free color-mode frequency as an observable quark or gluon mass. Running charged masses and mixing data must share scale and scheme [13, 14]. The finite source spectra of Section 3 have no certified running-to-pole, confinement, or laboratory-clock map, and they enter no row of the table below.

6.2 The twelve comparison values

The comparison branch is specified by a boundary packet: pixel coordinate $P = 1.630971465307864$, unified coupling $\alpha_U = 0.04112426184$, external unit $E_* = 1.22089 \times 10^{19}\,\mathrm{GeV}$, vacuum expectation value $v = 246.60010\,\mathrm{GeV}$ and scale $Q_0 = 91.58958701\,\mathrm{GeV}$. Its W and Z couplings use one-loop supersymmetric running, its Higgs and top values use imposed ultraviolet criticality, and its quark Yukawa couplings are the six-power texture

$$Y_u = \mathrm{diag}(6^{-6}, 6^{-3}, 1), \qquad Y_d = \mathrm{diag}(6^{-6}, 6^{-4}, 6^{-2}).$$

No proof here selects these values from finite repair. Multiloop outputs come from the pinned Standard Model in Dimensional Regularization code [12], with two corrections to its electroweak decoupling routine, the threshold-function operator and the placement of the strong-coupling assignment. A replay reproduces the five light quarks and the W, Z, Higgs and top to maximum relative discrepancy 5.85×10^{-11}. Table 3 gives the result against the 2026 Particle Data Group tables [16].

Table 3: The twelve nonzero comparison values, in GeV. All twelve lie outside their printed intervals. No covariance-aware joint significance or total theory-error bound is asserted.

	Output	Reference	Residual	Convention
e	0.000510962911	0.00051099895069	−70.53 ppm	Pole premise
μ	0.105650931342	0.1056583755	−70.45 ppm	Pole premise
τ	1.77680479525	1.77693	−70.46 ppm	Pole premise
u	0.00668144352	0.00216	+209.33%	$\overline{\mathrm{MS}}$, 2 GeV
d	0.00665065487	0.0047	+41.50%	$\overline{\mathrm{MS}}$, 2 GeV
s	0.239423082	0.0929	+157.72%	$\overline{\mathrm{MS}}$, 2 GeV
c	1.56844907	1.2729	+23.22%	$\overline{\mathrm{MS}}$ self-scale
b	6.59077908	4.186	+57.45%	$\overline{\mathrm{MS}}$ self-scale
t	170.611817	172.1	−0.865%	Pole, cross-section comparator
W	79.8563905	80.3625	−0.630%	Breit–Wigner
Z	90.5845168	91.1879	−0.662%	Breit–Wigner
H	124.842549	125.13	−0.230%	Pole

For positive outputs x_i and intervals $[l_i, u_i]$, one common factor is feasible exactly when $\max_i l_i/x_i \le \min_i u_i/x_i$. The intersection is empty for all twelve values, for the charged triple alone and for W, Z alone. This is an exact statement about the printed rectangle.

6.3 Charged leptons

The lepton circulant is chosen near the measured pole-mass Koide coordinates. Its response constants $50, 31, 310, 512, 77, 21, 27, 135$, its determinant exponent and the pole reading are target-informed: with the measured electron or muon mass the law returns $m_\tau = 1776.93\,\mathrm{MeV}$, within 2×10^{-3} of the world-average uncertainty, and this agreement carries no evidential weight. Absolute values use the external unit E_*. Summing the three cosines and their squares proves $Q = \sum_j m_j/(\sum_j \sqrt{m_j})^2 = (1 + e^{-2\chi})/3$, and the packet has $Q = 0.66666446$, $\rho = \sqrt{2}e^{-\chi} = 1.41420889$ and $\delta = 2/9 + \zeta = 0.22222476$, against the same coordinates of the measured pole triple to nine digits.

Reading the outputs as pole masses is an adopted modeling premise. Under it the discrepancy is almost entirely a common scale of about -70.5 parts per million. The muon/electron ratio residual is $+0.0725$ parts per million and the tau/electron ratio residual is $+0.0662$ parts per million. A common normalization cannot remove the first, and the electron and muon intervals admit no shared factor. The exact Koide point $\rho = \sqrt{2}$, $\delta = 2/9$ gives m_μ/m_e off by $+9.83$ parts per million and m_τ/m_e off by $+30.9$ parts per million; its tau ratio lies in the printed rectangle and its muon ratio does not. Assigning the same outputs running masses at Q_0 and applying the retained one-loop matching relation instead shifts the pole comparisons by $(4.854, 2.870, 1.820)$ percent. Proximity to pole coordinates selects neither scheme.

6.4 Quarks

The declared texture has the transported light ratios $m_s/m_d \simeq 35.99993$ and $m_u/m_d \simeq 1.00463$, in the modified minimal subtraction scheme with quantum chromodynamics and electrodynamics, four active quarks and three charged leptons at 2 GeV. The rectangle of the two quoted strange and down intervals is approximately $[19.33, 20.22]$, and the directly quoted m_u/m_d interval is $[0.442, 0.482]$. A common multiplier cancels in each ratio, so no energy unit can repair the light-quark shape, and this texture is rejected. Precise identification with the reference quantum-chromodynamic masses also needs an electromagnetic scheme conversion.

For a coefficient $m = A\,6^{-n}$ and ratio $r =$ output/reference, holding the transport factor fixed gives $n_{\mathrm{required}} = n + \log_6 r$. The u, d, s, c, b coordinates are approximately 6.63023, 6.19375, 4.52837, 3.11653 and 2.25334. Self-scale masses and threshold dependence require a coupled rerun for an actual changed boundary, so these are diagnostic coordinates and no selected replacement texture. Matched five-flavor running gives $\alpha_s(91.1879\,\mathrm{GeV}) = 0.11944066$ for the stipulated branch; a reference-ratio estimate of 0.11945742 is a different operation.

6.5 Weak bosons, Higgs and top

The W, Z, H, t residuals depend on the renormalization convention and attachment scale of the supplied vacuum expectation value. The source value agrees with the reference model's $v(172.4\,\mathrm{GeV}) = 246.60030\,\mathrm{GeV}$ within one part per million, while the branch attaches it at Q_0, where the reference value is 248.38564 GeV. Table 4 substitutes reference inputs at Q_0 to diagnose this sensitivity. A substitution derives no attachment, and the numerical equality supplies no scale-selection theorem. The controls omit the stored finite-light shifts of 1.7 and 13 keV.

The weak vectors use the stated Breit–Wigner convention, and the top comparison uses the $172.1 \pm 0.6\,\mathrm{GeV}$ cross-section pole extraction, distinct from a direct event-generator mass. Loop-order truncation, reference-input uncertainties and physical model error are distinct from numerical replay precision, and criticality itself is sensitive to loop order: at two loops the replayed boundary has $\lambda = 2.3 \times 10^{-8}$, and at three loops it has $\lambda = 9.7 \times 10^{-4}$.

Table 4: Percent residuals under reference substitutions at Q_0.

Inputs replaced by reference values	W	Z	H	t
None	-0.630	-0.662	-0.230	-0.865
Gauge couplings	-0.717	-0.698	-0.241	-0.899
v	$+0.076$	$+0.047$	$+0.436$	-0.200
v and gauge couplings	-0.012	$+0.011$	$+0.425$	-0.233
v, y_t, λ	$+0.063$	$+0.029$	$+0.110$	$+0.245$
All running inputs	-0.025	-0.008	$+0.099$	$+0.210$

6.6 Massless and neutral sectors

The photon and gluon zeros are conditional action coefficients. The minimal neutrino action gives three zero masses and zero oscillation gaps, contrary to the nonzero measured gaps, and individual absolute neutrino masses are unmeasured. The specified unshifted amplitude-to-mass attachment is excluded by Proposition 7.3. No particle abundance or early-Universe population is computed by this branch.

These are conditional comparisons, including target-informed choices. None of these constructions is a derivation of the measured spectrum. The finite gauge and matter results select no continuous mass spectrum, mixing matrices or population history. A joint physical test requires one source law, its state and complete record map, compatible mass conventions and declared experimental and theoretical uncertainties. Exact equality to measured central values is not an experimental theorem; equality of observable laws is the stronger mathematical target.

7 What records and symmetries exclude

Each statement below is exact on its stated domain and removes a tempting route to a spectrum. None excludes OPH as a whole; together they locate the missing selection.

7.1 Coherent phases and family averaging

The native event rates $p = 1/12$, $q = 1/6$ satisfy $2\sqrt{p/q} = \sqrt{2}$ and $p + q - 2pq = 2/9$, the base coordinates of the charged circulant.

Proposition 7.1 (Classical records select no coherent phase)**.** *For a finite instrument with Kraus operators K_{ra} and arbitrary real θ_{ra}, replacing K_{ra} by $e^{i\theta_{ra}}K_{ra}$ leaves every outcome map, every repeated composition and every record-dependent feedback experiment unchanged, on the full matrix space and with arbitrary entangled references. A well-defined function of the instrument maps therefore cannot depend on these phases.*

A coherent sum $\sum_{ra} c_{ra}K_{ra}$ thus does not follow from the instrument alone, and setting a probability equal to an angle in radians is an additional rule. An explicit same-rate counterfamily, separate from the native 128-record instrument, has identical classical instruments and different coherent spectra. A source-selected Stinespring isometry with an accessible coherent reference can carry the missing information, and such a reference is outside this exclusion.

Proposition 7.2 (Real cyclic averaging forces a repeated eigenvalue)**.** *For a real regular three-cycle*

R and $P = (I + R + R^2)/3$,

$$\tfrac{1}{3}\sum_{j=0}^{2} R^j X (R^T)^j = aP + b(I - P), \qquad X = X^T \in M_3(\mathbb{R}).$$

The two-dimensional rotation plane forces the repeat, so cyclic averaging of a real symmetric kernel cannot select three distinct family values. An equivariant construction from real inputs has real outputs, and a nonreal chiral response needs an input or selection rule supplying the oriented complex structure.

7.2 The passive family metric

For the native core-biased reference and nonnegative seam weights, stationarity gives $\sigma^T(I-T)\sigma = (157/90000)W_{\mathrm{cross}}$, which forces all eight core-to-outer weights to vanish. Every retained branch conserves a nonzero core-charge coordinate. The harmonic extension U of the native passive Schur calculation obeys $B^T U = GI$ and

$$Z = U^T U/G = I + (U - B)^T (U - B)/G \geq I.$$

The Schur mass matrix has the conserved family direction as a null vector, and the maximal 22-seam mixture attains rank two. Positive normalization is an invertible congruence and preserves rank, so it cannot generate the third charged family coefficient. A different reference or active recovery changes the premises of this obstruction.

7.3 Neutral hierarchy

Proposition 7.3 (Neutral squared-gap bound)**.** *Put $z = e^{-ih}$, $U = I+(z-1)P_2$, $V = I+(z-1)P_3$ on the bright plane of the two native rank-one family projectors, which have $\mathrm{tr}(P_2P_3) = 1/5$, and let $F = \left(\begin{smallmatrix} (1-p)U & U \\ pV & 0 \end{smallmatrix}\right)$ be the recurrent transfer. For $0 \leq p \leq 1/12$, $0 < h \leq 1/2$, its two roots of modulus above $1/2$ have phase coefficients $\omega, 1-\omega$ with*

$$0 \leq \omega \leq a = \tfrac{1}{2}\left(1 - \tfrac{95}{96}\sqrt{\tfrac{653}{848}}\right).$$

Under the common-scale attachment $(m_1, m_2, m_3) = M(0, \omega, 1-\omega)$, the adjacent squared-gap ratio is at most $a^2/(1-2a) < 0.004988$.

The determinant of F reduces to a four-dimensional calculation, with the other ten memory inputs retained as a transient block of induced norm at most p. Substituting $\lambda = e^{-ih/2}x$ and $y = x - p/x$ reduces the quartic to $y = (1-p)\cos(h/2) \pm i\sqrt{C_0}\sin(h/2)$, with $C_0 = (1-p)^2 + 4p/5$, and elementary bounds on the slow root give the estimate. The measured normal-ordering ratio is 0.03108, with printed rectangle $[0.03011, 0.03208]$, which excludes this attachment. Arbitrary neutral actions, offsets and other mass readouts are outside the proposition, and it fixes no absolute neutrino mass.

7.4 Residues, references and mass readouts

The complete instrument determines which spectral values an observer can read. Let

$$T = \begin{pmatrix} 1-p & 1 \\ p & 0 \end{pmatrix}, \qquad \pi = \frac{(1,p)}{1+p}, \qquad U_j = \exp(-ihX \otimes P_j),$$

with $0 \le p \le 1/12$. On the bright plane B, choose its antisymmetric area form J and put $A = |+\rangle\langle -| \otimes J$, $O = A + A^*$, $P_B = I_2 \otimes B$. Then $U_j A U_j^* = e^{-ih} A$ and $O^2 = P_B$. The three projectors $(P_B + O)/2$, $(P_B - O)/2$ and $I - P_B$ give a complete readout with values $+1, -1, 0$. Starting from π and the stationary bright, dark or tracial response state $P_B/4$, $(I - P_B)/2$ or $I_6/6$ gives, respectively,

$$C_B(n) = \cos(nh), \qquad C_D(n) = 0, \qquad C_{\mathrm{tr}}(n) = \tfrac{2}{3}\cos(nh).$$

The same transfer frequency therefore has a nonzero or a zero measured residue depending on the supplied state and detector, with the first measurement operation included.

Density records also cannot fix absolute field frequencies. Adjoin a reference line and extend each branch by $V_j(\delta) = 1 \oplus e^{-i\delta h} U_j$. All complete record probabilities and conditional density matrices confined to the original response sector are independent of δ, while coherences with the added reference acquire that phase, $F_\delta = e^{-i\delta h} F_0$.

A coherent family potential need not be a scalar mass. For the supplied Dirac symbol $\alpha_i = Z \otimes \sigma_i \otimes I$, a potential $I_4 \otimes A$ has zero projection onto the spin-invariant terms anticommuting with all α_i. For Hermitian A, M and $H_h(k) = \alpha \cdot k + X \otimes I_2 \otimes M + h I_4 \otimes A$,

$$\mathrm{Tr}[(H_h(k)^2 - |k|^2 I)^2] - \mathrm{Tr}[H_h(0)^4] = 16h^2 |k|^2 \,\mathrm{Tr}(A^2),$$

so a nonzero potential in this truncated Hamiltonian is no collection of constant relativistic masses. Finally, cyclic trace gives $\mathrm{Tr}([K, P]H) = \mathrm{Tr}(K[P, H])$. If $[P, H] = 0$, $[K, P] = iH$ and H is Hermitian, then $\mathrm{Tr}(H^2) = 0$ and $H = 0$. Exact global finite translation and boost identities therefore require a different domain from a nonzero relativistic Hamiltonian; controlled sectors and infinite limits carry their own error bounds.

7.5 Comparing capacities

Finite tests can compare whole record laws without enumerating histories.

Proposition 7.4 (Sufficient criterion for equal record laws)**.** *Let A_y, B_y be nonnegative outcome matrices of two finite classical instruments with column-stochastic sums, and C a nonnegative column-stochastic map. If $CA_y = B_y C$ for every record y and $Cp = q$, then all finite joint record laws coincide. With $\epsilon_0 = \|Cp - q\|_1$ and $\delta = \max_j \sum_{y,i} |(CA_y - B_y C)_{ij}|$, the record total-variation distance after n steps is at most* $\min(1, (\epsilon_0 + n\delta)/2)$.

Induction gives the exact case, and a telescoping sum of ℓ_1 contractions gives the approximate one. Equality of unread channels is insufficient: the one-state instruments $A_0 = 1, A_1 = 0$ and $B_0 = 0, B_1 = 1$ have the same unread channel and disjoint records. Determinism supplies neither the intertwiner nor a compatible initialization.

8 The catalogue and what a source-selected spectrum must supply

8.1 What the displayed source assigns

The common finite construction specifies three distinct charged-lepton coefficients, six quark coefficients with color multiplicities, and three zero neutrino coefficients in a 45-component internal kernel. The gauge and Higgs directions belong to the separate conditional bosonic action.

Channel	Displayed source input	Established finite meaning
Charged leptons	$Y_e = \sqrt{2}\, g Z_C$	Three memory-derived singular coefficients on the native charged-lepton slots
Up and down flavors	Diagonal powers of 1/6	Six declared coefficients on native color slots; this displayed choice has identity quark mixing
Minimal neutrino slots	Zero	Three zero internal coefficients, without a massless relativistic particle proof
Gauge and scalar fields	Canonical one-doublet action, g_i, λ_H, v	Conditional symmetry breaking and quadratic masses; no bosonic defect-state inclusion in the CAR construction

This is a complete accounting of the displayed model's assignments, not a complete catalogue of nature. Two mathematical distinctions prevent a false physical spectrum. First, the compact four-mode band has a nonzero operator $A = W^\dagger h W$, whose energies enter the exact common operator and cannot be removed to retain the zero-mode mass list. Second, the positive quartic coupling changes the spectral poles. The pixel feedback calculation uses quadratic coefficients, so the compact interacting operator and its exact slice readouts require a recalculated electromagnetic response before a common interacting feedback fixed point can be claimed.

8.2 The required attachments

Table 5 states what each missing attachment must deliver. Replacing any of these by measured values calibrates a model; an untuned spectrum identity requires the listed derivation.

Table 5: Attachments between the finite stack and a physical spectrum.

Attachment	**Required result**
Charged-family source law	Physical scalar Yukawa eigenvalues, phase and flavor frames selected without measured-target fitting
Gauge, Higgs and top parameters	Gauge content, running and matching prescription, vacuum scale and the criticality premise derived from the same source
Neutral masses and mixing	A nonzero-gap mechanism and complete readout outside the excluded minimal and unshifted-phase branches
Physical records	Matched preparation, instruments, clock and estimator, retaining all failure outcomes, with a complete error budget
Capacity and refinement	A full-history equivalence or controlled limit for the physical source, beyond the four-address encoding
Early-Universe distribution	The relevant state, finite-temperature dynamics, interactions and population observables from the same source

These combine into one joint map,

(same source, vacuum, clocks, instruments)
$\longrightarrow$ (total defect energies, current residues, dispersion, vertices)
$\longrightarrow$ (one response kernel and its pixel fixed point).

Its first row must supply an autonomous finite-energy defect process in place of an externally fixed texture. Its second must retain all spatial regulator modes until an actual decoupling or continuum theorem controls them. The third must use the first two, including thresholds and physical normalization. These are composition requirements on one permitted design; uniqueness among all possible simulator designs is not a prerequisite.

8.3 Interaction, chiral continuum and responsive geometry

The quartic result gives a connected interaction on a fixed localized algebra, with a bound uniform in the surrounding radial volume. Its spatial tails and fixed number of localized modes do not define an interacting local $3+1$ field theory. A controlled common field limit must also establish the intended chiral measure, anomaly realization, local gauge constraints, surviving interactions and regulator census. Finite internal anomaly sums are necessary representation checks; a chiral lattice measure has additional locality and integrability requirements [19, 20, 21].

The same matter energy and stress must then enter the geometric response. The companion Einstein implication uses compatible clocks and metric, conserved stress, a geometric modular normalization and controlled fixed-proper-volume generalized-entropy variations [4]. The finite Gibbs first law proved for the interacting localized Hamiltonian is a common-operator identity, and it supplies no geometric area term, small-ball remainder, Ward/Bianchi map or universal Newton normalization.

Reproduction and mathematical scope

Every statement in this paper has a public executable certificate, proof text or formal module in the project repository [1], including exact characteristic polynomials, rational and integer enclosures, complete line tables, adversarial controls and the source-pinned multiloop replay. They distinguish exact algebra and rational interval proofs from floating spectral checks and empirical comparisons. A source hash identifies a calculation; its mathematical scope is the stated theorem and domain.

References

[1] B. Müller et al., *Observer Patch Holography: formal proofs and executable receipts*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/main/model

[2] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, and D. A. Friedman, *Observers Are All You Need*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/observers_are_all_you_need.pdf

[3] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/reality_as_consensus_protocol.pdf

[4] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[5] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, D. Matscheko, and D. A. Friedman, *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[6] B. Müller, M. Poneder, J. Kim, and D. A. Friedman, *An Executable Observer for Observer Patch Holography: Specification, Memory and Observable Fields*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/executable_observer_construction.pdf

[7] Y. Koide, *New view of quark and lepton mass hierarchy*, Physical Review D **28**, 252 (1983).

[8] C. A. Brannen, *The Lepton Masses*, preprint, 2006. http://brannenworks.com/MASSES2.pdf

[9] H. Georgi and S. L. Glashow, *Unity of all elementary-particle forces*, Physical Review Letters **32**, 438–441 (1974).

[10] E. H. Lieb, *Two theorems on the Hubbard model*, Physical Review Letters **62**, 1201–1204 (1989).

[11] C. N. Yang, *η pairing and off-diagonal long-range order in a Hubbard model*, Physical Review Letters **63**, 2144–2147 (1989).

[12] S. P. Martin and D. G. Robertson, *Standard Model parameters in the tadpole-free pure* $\overline{\text{MS}}$ *scheme*, Physical Review D **100**, 073004 (2019), arXiv:1907.02500.

[13] S. Antusch, K. Hinze, and S. Saad, *Updated Running Quark and Lepton Parameters at Various Scales*, arXiv:2510.01312, 2025. https://arxiv.org/abs/2510.01312

[14] Y. Aoki et al. (Flavour Lattice Averaging Group), *FLAG Review 2024*, arXiv:2411.04268, 2024, Tables 11–12. https://arxiv.org/abs/2411.04268

[15] Particle Data Group, *Resonances*, Review of Particle Physics, 2025 update. https://pdg.lbl.gov/2025/reviews/rpp2025-rev-resonances.pdf

[16] Particle Data Group, *Review of Particle Physics: Summary Tables*, 2026. https://pdg.lbl.gov/2026/tables/rpp2026-sum-leptons.pdf; https://pdg.lbl.gov/2026/tables/rpp2026-sum-quarks.pdf; https://pdg.lbl.gov/2026/tables/rpp2026-sum-gauge-higgs-bosons.pdf.

[17] D. Buchholz, *Gauss' law and the infraparticle problem*, Physics Letters B **174**, 331–334 (1986).

[18] P. Duch and W. Dybalski, *Infrared problem in quantum electrodynamics*, arXiv:2307.06114.

[19] M. Lüscher, *Abelian chiral gauge theories on the lattice with exact gauge invariance*, Nuclear Physics B **549**, 295–334 (1999), arXiv:hep-lat/9811032.

[20] M. Lüscher, *Lattice regularization of chiral gauge theories to all orders of perturbation theory*, Journal of High Energy Physics **06** (2000) 028, arXiv:hep-lat/0006014.

[21] D. Kadoh and Y. Kikukawa, *A simple construction of fermion measure term in $U(1)$ chiral lattice gauge theories with exact gauge invariance*, Journal of High Energy Physics **02** (2008) 063, arXiv:0709.3658.

[22] Jarlskog, C. (1985). "Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation." *Physical Review Letters* **55**, 1039–1042. https://doi.org/10.1103/PhysRevLett.55.1039

[23] Peskin, M. E., and Schroeder, D. V. (1995). *An Introduction to Quantum Field Theory*. Addison-Wesley. https://doi.org/10.1201/9780429503559

[24] Weinberg, S. (1967). "A model of leptons." *Physical Review Letters* **19**, 1264–1266. https://doi.org/10.1103/PhysRevLett.19.1264

[25] B. Müller et al., *OPH basis and conditional theorem-premise registry*. https://github.com/FloatingPragma/observer-patch-holography/blob/main/claims/axiom_registry.yaml

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.